EXP.NO: 4	**Create a web form that accepts a user's name and age. Write a Servlet to process the form data and display it back on the browser.**
DATE:26/02/2025	

AIM:

To create a simple web application that includes a form to accept a user's name and age, and then use a Servlet to process the submitted data. The Servlet will retrieve the form data, process it, and display the values back on the browser.

ALGORITHM:

1. **Input:**
   - The user fills out a web form with their **name** and **age** and submits it.
2. **Form Handling and Servlet Processing:**
   1. **Create the Web Form:**
      - Create an HTML form with **two input fields**: one for the user's **name** and one for the user's **age**.
      - Add a **submit button** to the form.
      - Set the form's action attribute to the URL of the **Servlet** and the method to POST (or GET).
   2. **Create the Servlet Class:**
      - Write a Servlet that processes the **form data** submitted by the user.
      - The Servlet will override the doPost() (or doGet() depending on the form method) to handle form submissions.
      - The Servlet will **retrieve the name and age** parameters from the request object using request.getParameter("name") and request.getParameter("age").

3. **Process the Data in the Servlet:**
    - Extract the name and age values submitted from the form.
    - **Validate the input** (e.g., check if the age is a valid number).
    - Prepare a **response** that will display the user's **name** and **age** back on the browser.
4. **Send Response to the Browser:**
    - The Servlet will send a **response** to the browser containing a message like:
        - "Hello, [name]! You are [age] years old."
    - The response will be displayed to the user in the browser.
5. **Deploy the Servlet:**
    - Compile the Servlet and deploy it to a **Servlet container** (like **Apache Tomcat**).
    - Ensure the web.xml file or Servlet annotations are used to map the Servlet to the URL defined in the form's action attribute.
6. **Test the Application:**
    - Open the web browser, fill in the form with the name and age, and submit it.
    - The Servlet will process the data and display the message back to the user in the browser.

Program:

Form.html:

```
<!DOCTYPE html>
<html>
```

```
<head>
  <title>Submit </title>
  <style>
    body {
      font-family: Arial;
      background-color: #f2fcff;
    }

    .container {
      width: 400px;
      margin: 80px auto;
      background: white;
      padding: 20px;
      border-radius: 10px;
      box-shadow: 0 0 10px #ccc;
    }

    h2 {
      text-align: center;
      color: #004aad;
    }

    input, textarea {
      width:  100%;
      padding: 8px;
```

```
        margin: 8px 0;
      }

      button {
        width: 100%;
        padding: 10px;
        background: #004aad;
        color: white;
        border: none;
      }
    </style>
  </head>

  <body>

  <div class="container">
    <h2>Select a book</h2>

    <form action="servelt1" method="post">
      <input type="text" name="name" placeholder="Student Name" required>
      <input type="number" name="age" placeholder="Age" required>
      <input type="text" name="dept" placeholder="Department" required>
      <input type="text" name="roll" placeholder="Roll Number" required>
      <textarea name="message" placeholder="Enter Notice"></textarea>
```

```
    <button type="submit">Submit</button>
  </form>
</div>

</body>
</html>
```

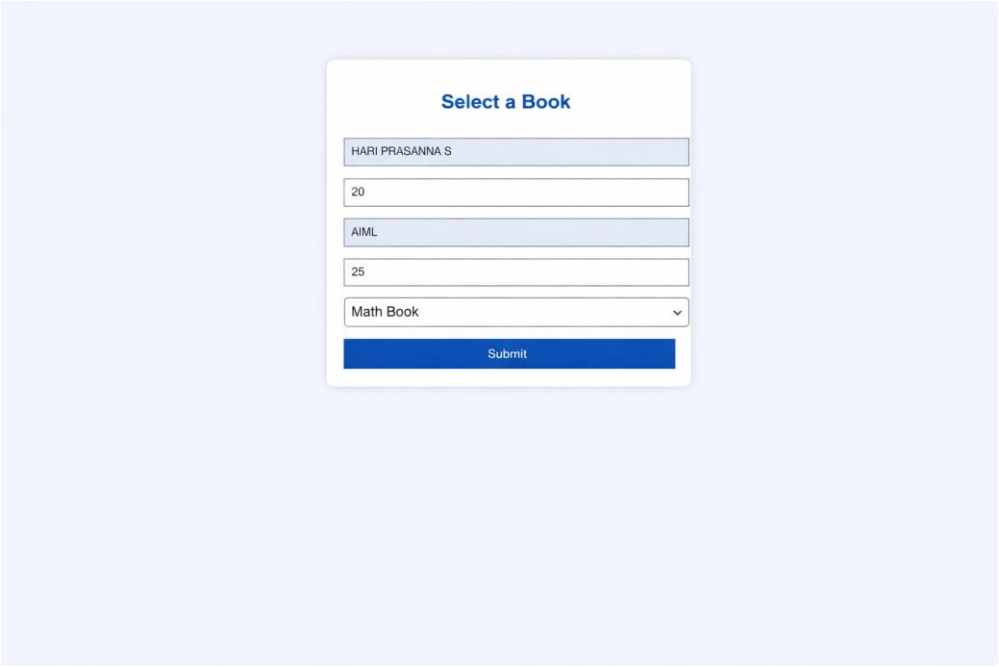
PostServlet.java:

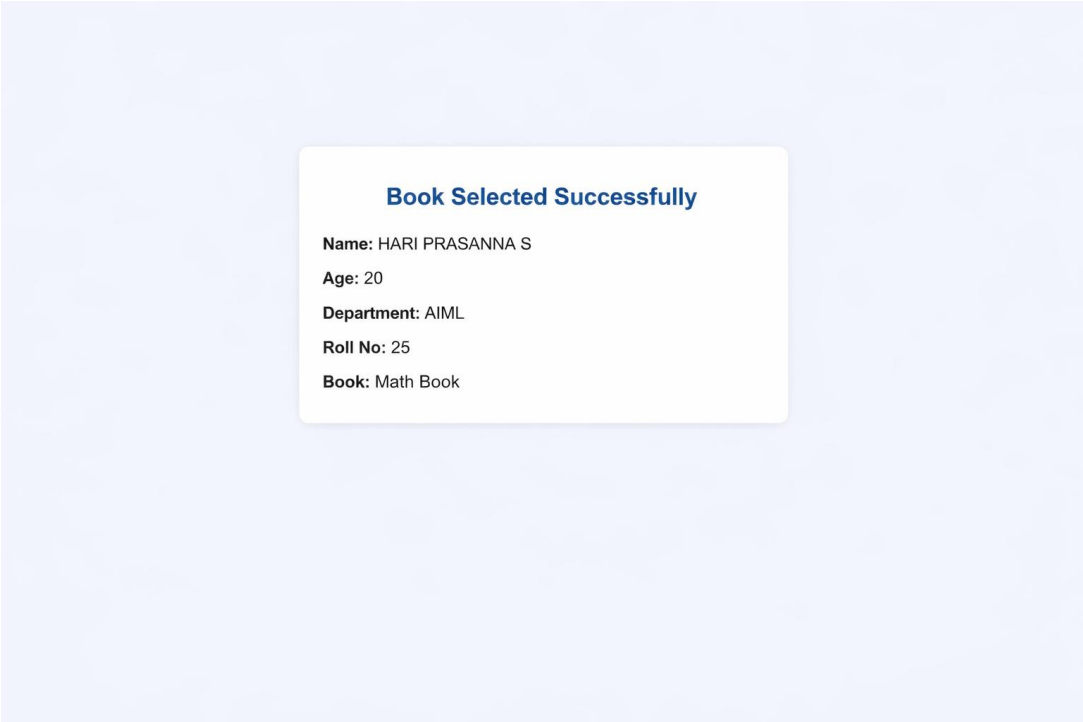
```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/servelt1")
public class servelt1 extends HttpServlet {
        protected void doGet(HttpServletRequest request,
HttpServletResponse response)
              throws ServletException, IOException {

           doPost(request, response); // redirect GET to POST
        }

  protected void doPost(HttpServletRequest request, HttpServletResponse
response)
```

```
    throws ServletException, IOException {

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    String name = request.getParameter("name");
    String age = request.getParameter("age");
    String dept = request.getParameter("dept");
    String roll = request.getParameter("roll");
    String message = request.getParameter("message");

    out.println("<html><body style='font-
family:Arial;background:#f2f6ff;'>");

    out.println("<div style='width:500px;margin:50px
auto;background:white;padding:20px;border-radius:10px;'>");
    out.println("<h2 style='color:#004aad;'>Notice Submitted
Successfully</h2>");

    out.println("<p><b>Name:</b> " + name + "</p>");
    out.println("<p><b>Age:</b> " + age + "</p>");
    out.println("<p><b>Department:</b> " + dept + "</p>");
    out.println("<p><b>Roll No:</b> " + roll + "</p>");
    out.println("<p><b>Notice:</b> " + message + "</p>");

    out.println("</div>");
```

```
        out.println("</body></html>");
    }
}
```

Output:





RESULT:

A Servlet is created to process the HTML form data and display it back on the browser.